

Alessandro Porcheddu

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Education

Technical University of Denmark

M.Sc. IN COMPUTER SCIENCE AND ENGINEERING

Copenhagen, Denmark

Sept. 2026 - Ongoing

- **Specialization:** Embedded Systems and IoT, Hardware/Software Codesign

University of Bologna

B.Sc. IN COMPUTER SCIENCE AND ENGINEERING

Cesena, Italy

Sept. 2023 - Jul. 2026

- **B.Sc. Thesis:** "Design and Development of a Software-Defined ECU with a Server-Side Digital Twin for Single-Cylinder Internal Combustion Engines" (C++, OOP, FreeRTOS, OTA, HITL, AFSM, Digital Twin, Custom WebApp, PCB Design)
- **Relevant Coursework (4.0 GPA):** Algorithms and Data Structures (**C**), Object-Oriented Programming (**Java**), Linear Algebra for Machine Learning (**Python**), Programming (**C**), Telecommunications Networks, Embedded Systems and IoT (**C**), Data Intensive Applications (**Python**)

Work Experience

UniBo Motorsport - Motostudent Competition

E-POWERTRAIN DIVISION MEMBER

Bologna, Italy

Sept. 2025 - Ongoing

- Designing and prototyping a **custom ECU** for motorcycle powertrain management and telemetry.
- Implementation of a specialized **AI Model** in the STM32 environment, inside the FreeRTOS system.
- Contributed to **PCB design** and component selection for the custom ECU hardware.

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SYSTEM ADMINISTRATOR INTERN

Cesena, Italy

Sept. 2025 - Dec. 2025

- Configured and maintained Windows **Domain Controller**, ensuring reliable system performance and uptime.
- Documented system configurations, network layouts, and **maintenance procedures** for internal use.
- Worked with virtualization tools such as **VMware and Proxmox** to create and test system environments.

Cables

AUTOMATION TECHNICIAN

Cesena, Italy

Jun. 2023 - Jun. 2023

- Assisted in assembling electrical control panels for **industrial automation systems**, following detailed wiring diagrams and specifications.
- Troubleshoot **PLC** input/output issues and performed routine checks on control system hardware.

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IT ASSISTANT

Wrocław, Poland

Jun. 2022 - Jul. 2022

- Completed the **Erasmus+** "Trainee for Industry 4.0 Plus" internship program, gaining practical experience in industrial environments.
- Assisted with the digitization of paper records by transferring them to solid-state storage systems.

Side Projects

JavaDyno (Engine Dynamometer Management Software)

[Code](#)

SOFTWARE DEVELOPMENT

- Developed an engine dynamometer management system (EDMS) with an **MVC** core structure using **JUnit** automated testing and **JavaFX**.
- Ensured **OBD-II** over **CAN** compatibility for in-vehicle testing, enhancing real-world applicability.
- Integrated **WebSocket** and **Serial** connection support for versatile device communication and included simulation features.

Motorcycle Parts E-commerce Platform

[Live](#)

WEB DEVELOPMENT

- Built a full-stack e-commerce platform using **Laravel 12**, **React 19**, and **Inertia.js**.
- Implemented secure payment processing with **Stripe** via **Laravel Cashier** and webhook signature validation.
- Developed an **admin dashboard** for product, order, and customer management with analytics and reporting tools.

Motorcycle Exhaust Valve Control Unit

[Code](#)

EMBEDDED SYSTEMS

- Developed an electronic exhaust valve control unit based on **ESP32**.
- Designed a custom PCB using **KiCad** and a custom 3D printed enclosure using **Fusion360**.

Skills

Technologies Java, C, C++, TypeScript, React, Python, PHP, Cuda, AWS, KiCad, Fusion360, Matlab, , CAN-Bus, IoT, JTAG, RTOS, VMware
Languages Italian, English